

Electronic Health Records



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Outline



- Lecture Learning Outcomes
- Introduction & Definitions
- Electronic Health Record
- Why do we need Electronic Health Records?
- Few Electronic Health Record Key Components
- Electronic Health Record Examples
- Barriers to Electronic Health Record Adoption

Lecture Learning Outcomes



- Outline the features and components of electronic health records
- Describe the role of electronic health records in clinical practice

INTRODUCTION

Definition of the term Electronic Health Record (EHR)

- There is no universally accepted definition of an EHR.
- EHRs are also known as electronic medical records (EMRs), computerized medical records (CMRs), electronic clinical information systems (ECIS) and computerized patient records (CPRs).





David (8 month and 10 day)
 (Males 0 years and 2 month)

Mother: Teacher
 Father: Financial advisor
 Parents: Married

Last Anderson P

First David Boy

Birth 5 January 2009

Age: 8 month and 10 days

Patient nb: 3

Appointments

Forms

Meeting (Doctor)
 Full status (Doctor)
 Assistant
 Billing
 Reports
 Statistics

Sheets

O: Neurologic
 O: Vascular
 O: Cardiac
 O: Respiratory
 O: Abdomen
 Exams
 Radiology
 Summary
 Patient documents
 Letter

SOP

Sum

T

R-V

T, P, PC

Admission

Agenda

Meetings

2 month checkup	5 Mar 09	2m, 0d
1 month checkup	5 Feb 09	1m, 0d
Respiration problem	22 Jan 09	17d
10 days checkup	13 Jan 09	9d
Control for return at home	9 Jan 09	4d
Birth	5 Jan 09	0d

Diagnosis

General
 My Diagnosis
 Social

New documents

Abdomen palpat
 15 Sep 2009
 Cardiac auscul
 15 Sep 2009

To Do

Send checkup

Area: 1

Doc: 8

Notes

Father ask many questions, add 10 minutes to consultation

Current doctor Dr Herran

Menu 1

Menu 2

Menu 3

Search

Digestive

Digestive inspection

Normal

Digestive auscultation

Normal abdomen noises

Digestive palpation

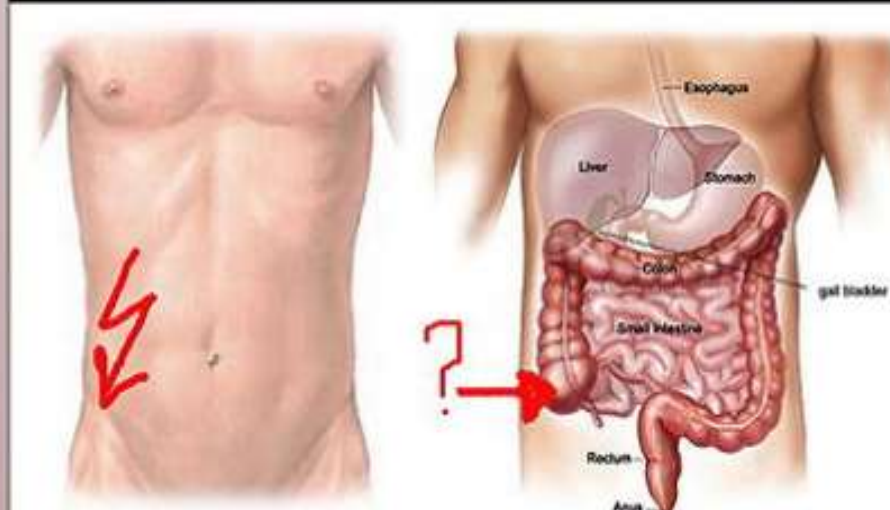
Little pain on the right lower area

Thursday, 22 Jan 2009

Liver

No hepaticomegaly.

Rectal



Page 1/1

Draw ☒

Mark

Color

Pen

8

Documents
 manager



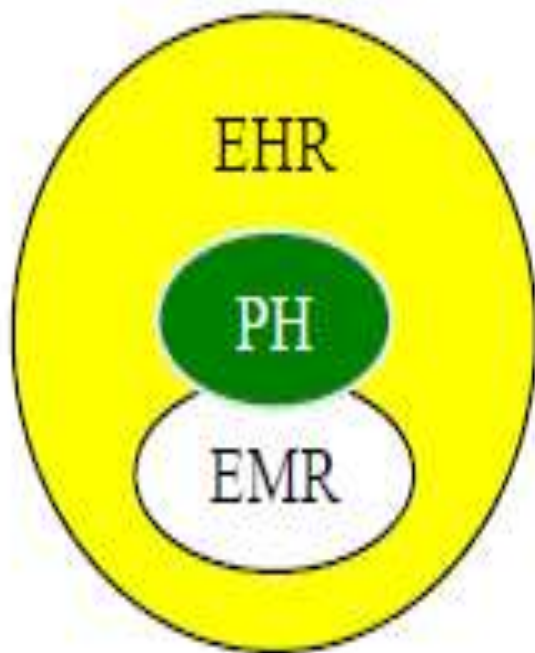
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Relationship between EHR, EMR and PHR



- The EHR is the larger system that includes the EMR and PHR and interfaces with multiple other electronic systems locally, regionally and nationally.
- The EMR, on the other hand, is the electronic patient record located in an office or hospital
- The PHR is a collection of health information by and for the patient.
- There is overlap between the EMR and the PHR, since the PHR can be part of the EMR.



Relationship between EHR (electronic health record), PHR (personal health record) and EMR (electronic medical record)

Electronic Medical Record (EMR)



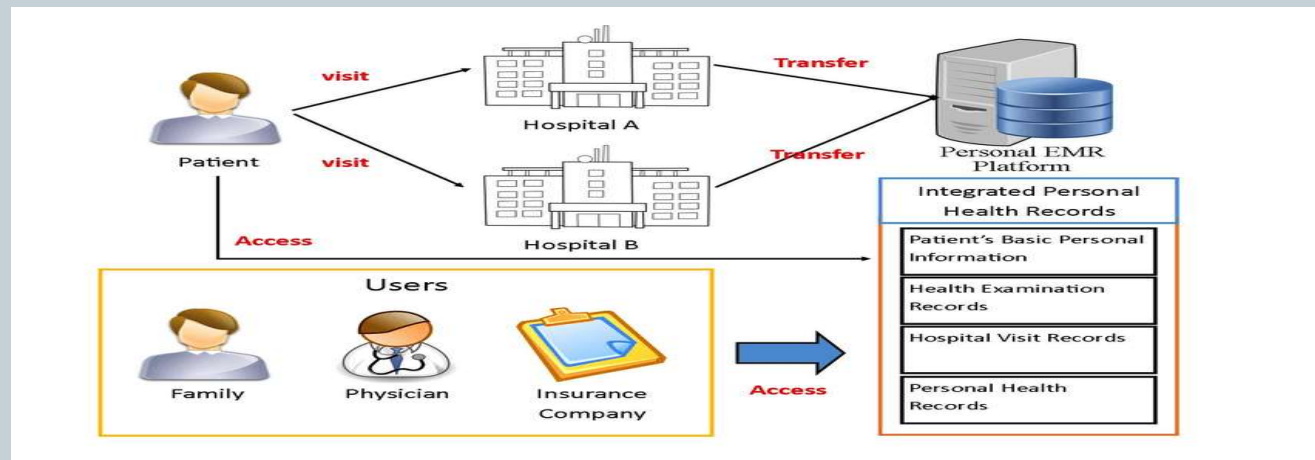
- “An electronic record of health-related information on an individual that can be created, gathered, managed and consulted by authorized clinicians and staff within one healthcare organization”.



Personal Health Record (PHR)



- “An electronic record of health-related information on an individual that conforms to nationally recognized interoperability standards and that can be drawn from multiple sources while being managed, shared and controlled by the individual”.



Electronic Health Record



- “An electronic record of health-related information on an individual that conforms to nationally recognized interoperability standards and that can be created, managed and consulted by authorized clinicians and staff across more than one healthcare organization”.



Types of EHR



Electronic Health Record

- Outpatient (ambulatory)
- Inpatient

Why do we need Electronic Health Records?



- The paper record is severely limited.
- The need for improved efficiency and productivity.
- Quality of care and patient safety.
- Public expectations.
- Governmental expectations.
- Financial savings.
- Technological Advances.
- Older and more complicated patients require more coordinated care.

1. Limitations of Paper Records



- Illegible handwriting
- Cannot be electronically shared or stored
- Not structured data
- Expensive to copy, transport and store
- Easy to destroy
- Difficult to analyze and determine who has seen it
- The negative impact on the environment
- Cannot rapidly retrieve information
- Chances of missing



Figure 2.2 Outpatient paper-based patient encounter form

2.The need for improved efficiency and productivity



With an EHR:

- lab results can be retrieved much more rapidly, thus saving time and money. Reducing duplicated tests and prescriptions.
- EHRs are more efficient because they reduce redundant paperwork and have the capability of interfacing with a billing program that submits claims electronically.
- No playing “telephone tag”.
- Help with productivity if templates are used judiciously
- Embedded educational content for clinicians is one of the newest features of a comprehensive EHR.
- Centralized area for all physician approvals and signatures of lab work, prescriptions, etc



Home | Bowers, George | Allergies: Tylenol 3 | PCP: (None) | Alerts: MEDICARE | MyChart: Inactive

Activities | Clinical Distinctions | Master Index | Group Search

Relevant Documents | Additional Search

Diagnosis/Problem Related Documents

- Angina (Disorder)
- Acute Chest Pain (DDx, Adolescent)
- Angina (Patient Handout)

Document Preview | Patient Instructions

Back | Forward | Master Index

Angina | FIRSTConsult

Summary | Diagnose | Treatment | Outcomes | Prevention

Summary | Therapies | Signs and Symptoms

Summary of therapies

- Medical management is the mainstay of treatment, often controlling angina satisfactorily for many years.
- Catheter interventions and coronary bypass surgery are treatments, not cures, and underlying issues of risk factor management must still be addressed.
- The use of nitroglycerin prior to physical activity can often prevent angina and should be recommended to active patients with predictable angina.

In treatment of patients with angina, the following four strategies should be kept in mind:

Pharmacologic management

Selected Documents | Add to instructions

One click of a button looks up the evidence-based treatment and patient management information relevant to this patient. No searching or log-in needed!

Figure 2.3 iConsult integrated with Epic EHR (courtesy iConsult)

3. Quality of care and patient safety

- Improve patient safety through many mechanisms: (a) Improved legibility of clinical notes, (b) Improved access anytime and anywhere (c) Reduced duplication (d) Reminders that tests or preventive services are overdue (e) Clinical decision support that reminds us of patient allergies, the correct dosage of drugs, etc. (f) Electronic problem summary lists provide diagnoses, allergies and surgeries at a glance.



Good
legibility

POOR
LEGIBILITY

Good readability.
More comfortable
to read.

Poor Readability.
Less comfortable
to read.

4. Public expectations

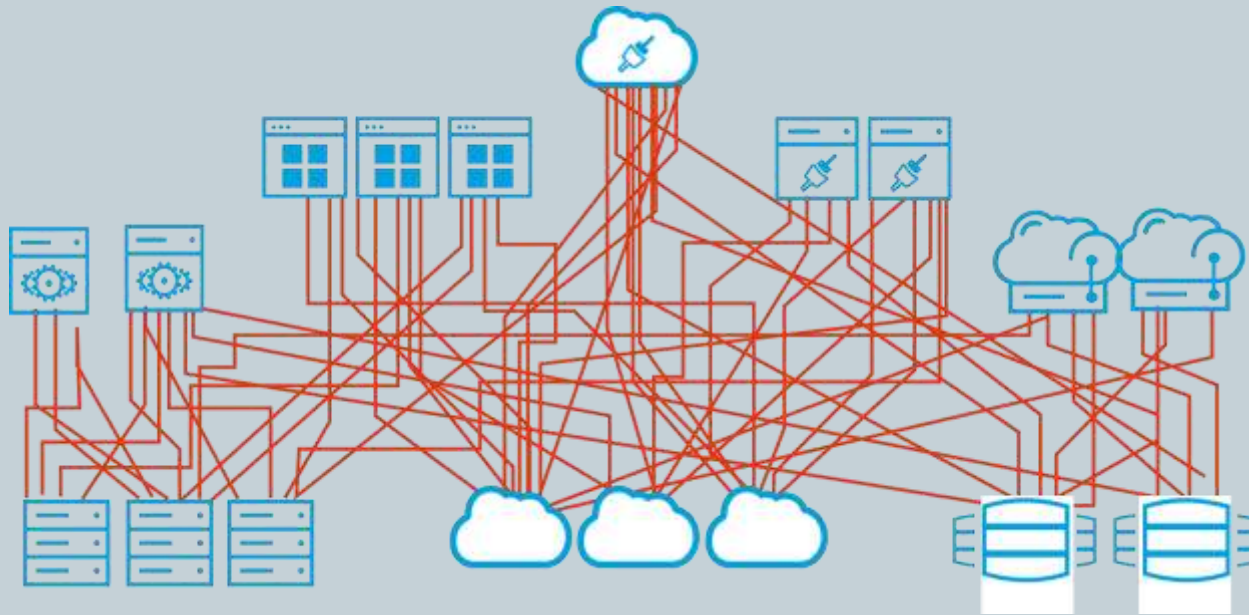


- Decrease medical errors
- Reduce healthcare costs
- Influence decision about selecting a personal physician.
- Better customer satisfaction through fewer lost charts, faster refills and improved delivery of patient educational material.



5. Governmental Expectations

- An interoperable electronic health record
- Coordinate disease management in older patients.



6. Financial savings



- Eliminated chart rooms and record clerks
- Reduction in the need for transcription
- Fewer callbacks from pharmacists with electronic prescribing
- Copying expenses and labor costs would be reduced
- More efficient patient encounters mean more patients could be seen each day.
- Decrease malpractice



7. Technological Advances..



- Application service provider (ASP) concept
- Computer speed, memory and bandwidth have advanced such that digital imaging is also a reality, so images can be part of an EHR system.
- Standard PCs, laptops and tablet PCs continue to add features and improve speed and memory while purchase costs drop.
- Wireless and mobile technologies permit access to the hospital information system, the electronic health record and the Internet using a personal digital assistant, smart phone or laptop computer.



8.Older and more complicated patients require more coordinated care...



Common for older patients to have more than one physician.

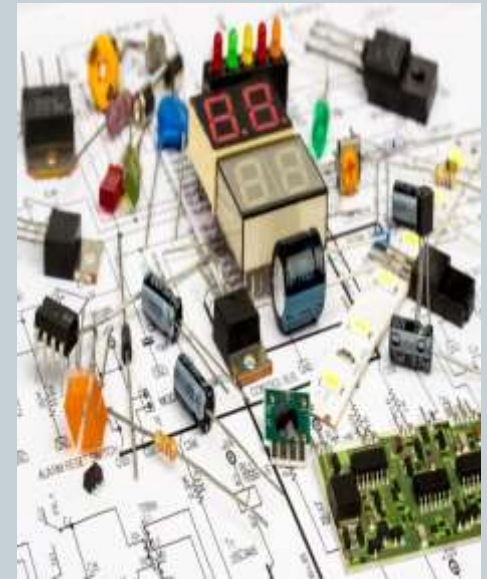
EHR will;

- Improving coordination care
- Improving communication between disparate healthcare entities.

Few Electronic Health Record Key Components

The following components are desirable in any EHR system:

- Clinical Decision Support Systems (CDSS)
- Secure messaging (e-mail)
- An interface with practice management software, scheduling software and patient portal
- Referral management feature
- Retrieval of lab and x-ray reports electronically
- Retrieval of prior encounters and medication history
- Computerized Physician Order Entry (CPOE)
- Electronic patient encounter



Computerized Physician Order Entry (CPOE)

- CPOE is an EHR feature that processes orders for medications, lab tests, x-rays, consults and other diagnostic tests.

The screenshot shows a 'Medication Order' window for 'WARFARIN TAB'. It includes a 'Change' button, a warning 'CAUTION IN WOMEN OF CHILD-BEARING YEARS* - SOUND-ALIKE/LOOK-ALIKE DRUG', and a table with columns: Dosage, Complex, Route, and Schedule. The table lists dosages from 10MG to 12MG with corresponding complex values. The 'Route' column has 'ORAL' selected. The 'Schedule' column has 'QDAILY' selected. Below the table is a 'Comments' field with the text 'Take 7.5 mg a day on Mon-Fri only'. There are also fields for 'Days Supply' (30), 'Qty (TAB)' (30), and 'Refills' (7). A 'Pick Up' section has radio buttons for 'Clinic', 'Mail', and 'Window', with 'Mail' selected. A 'Priority' dropdown is set to 'ROUTINE'. A 'Patient Instruction' section has a checkbox for 'TO PREVENT CLOTS' and a text box with 'CDI "HIGH ALERT MEDICATION"'. At the bottom, there is a summary line: 'WARFARIN TAB 10MG TAKE ONE TABLET BY MOUTH EVERY DAY TO PREVENT CLOTS take 5 mg a day Sat & Sun Quantity: 30 Refills: 7'. There are 'Accept Order' and 'Quit' buttons. Red arrows point to the 'Comments' field with the annotation 'However, this can lead to inconsistencies' and to the 'Comments' field with the annotation 'Flexibility allows for clarification through a free-text comment window'.

Dosage	Complex	Route	Schedule
10MG	0.1217	ORAL	QDAILY
2MG	0.1251	ORAL	QAM AND NOON
2.5MG	0.1314		QAM AND QHS
3MG	0.2434		QAM AND NOON QHS
4MG	0.1273		QAM AND QPM
5MG	0.1878		QDAILY
6MG	0.1921		QHS
7.5MG	0.1816		QID
10MG	0.3756		QID TMT
12MG			QMONTHLY

Comments: Take 7.5 mg a day on Mon-Fri only

Days Supply: 30 Qty (TAB): 30 Refills: 7

Pick Up: ☐ Clinic ☒ Mail ☐ Window

Priority: ROUTINE

Patient Instruction: ☒ TO PREVENT CLOTS

CDI "HIGH ALERT MEDICATION"

WARFARIN TAB 10MG
TAKE ONE TABLET BY MOUTH EVERY DAY TO PREVENT CLOTS take 5 mg a day Sat & Sun
Quantity: 30 Refills: 7

CPOE will:

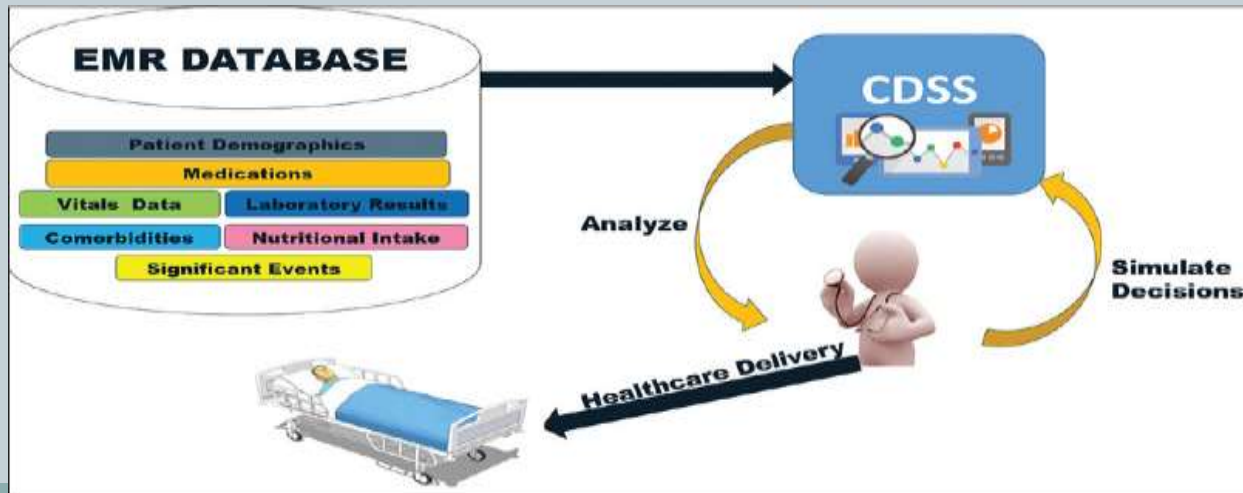


- Reduce medication errors.
- Reduce costs.
- Reduce variation of care.

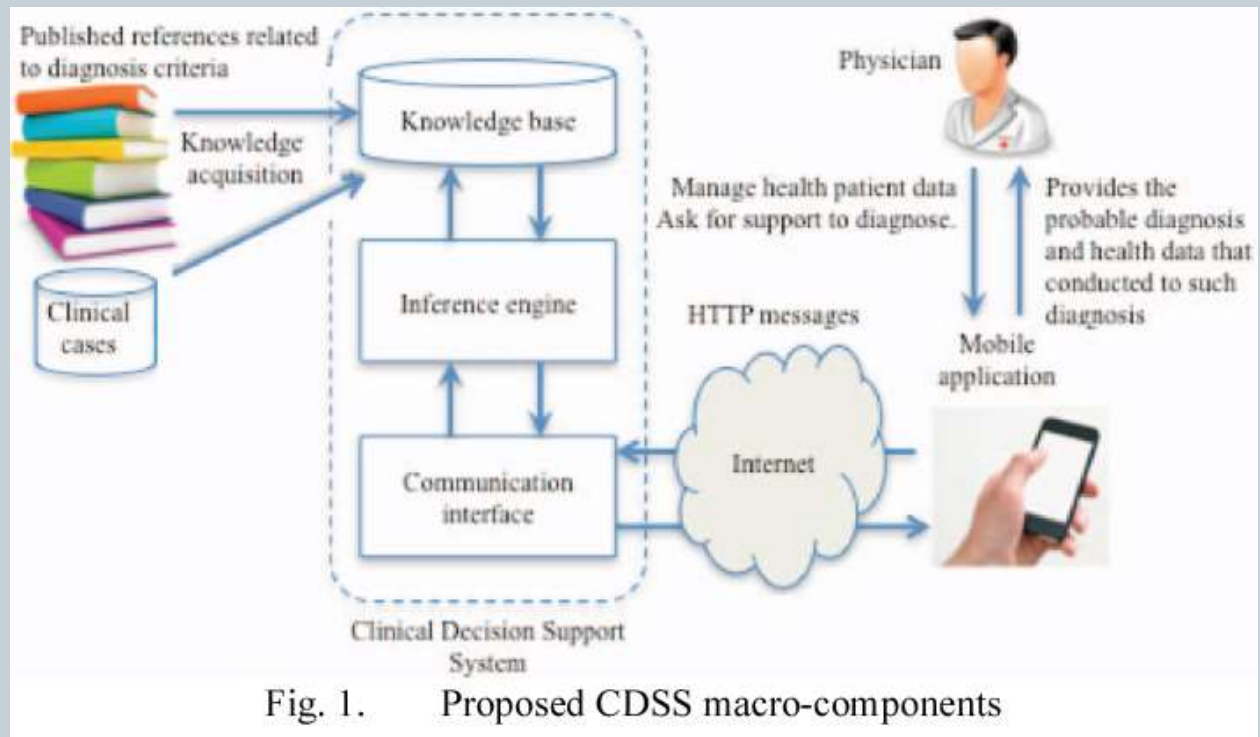
Clinical Decision Support Systems (CDSS)



- Definition: “any software designed to directly aid in clinical decision making in which characteristics of individual patients are matched to a computerized knowledge base for the purpose of generating patient specific assessments or recommendations that are then presented to clinicians for consideration”



- Most EHR systems may offer alternatives and provide reminders but make no decisions on their own.



Some of the clinical decision support.....



Table 2.5 Clinical decision support

Type of CDSS	Examples
Knowledge	iConsult®, Theradoc®
Trending/ Patient tracking	Flow sheets graphs
Medications	CPOE and drug alerts
Order sets/protocols	CPGS and order sets
Reminders	Mammogram due
Differential diagnosis	Dxplain®
Radiology CDSS	What imaging studies to order?
Laboratory CDSS	What lab tests to order

Electronic Health Record Examples



1. Open-Source and Free EHRs: OpenVista, WorldVista, Medsphere OpenVista etc.
2. Low Cost EHR Programs: Medical Office Online, SoapWare etc.
3. More Comprehensive EHR: Armed Forces Health Longitudinal Technology Application (AHLTA), Veterans Health Administration Vista etc.

Barriers to Electronic Health Record Adoption



- Historic constraints to EHR adoption:
 - 1) The need for standardized clinical terminology;
 - 2) Privacy, confidentiality and security concerns;
 - 3) Challenges to data entry by physicians and
 - 4) Difficulties with integrating with other systems



What are the other barriers?



- Financial Barriers.
- Physician resistance.
- Loss of productivity.
- Work flow changes.
- Integration with other systems.
- Lack of standards.
- Adverse legislation.
- Inadequate proof of benefit.

References



- Chapter 2, Medical Informatics: Practical Guide for the Healthcare Professional , 3rd Edition



Thank You